
Dealer Pays:	Fixed
Client Pays:	1-year LIBOR
Notional:	\$100mm
Maturity:	4 Years
Payment Frequency:	Annual, on both sides
Day Count:	30/360, on both sides
Knockout Feature:	If, on any reset date, 1-year LIBOR is less than or equal to 4.75%, then there will be no payments on either side for that period

Table 6.6: Terms of a Knockout Swap in Our Stylized Example

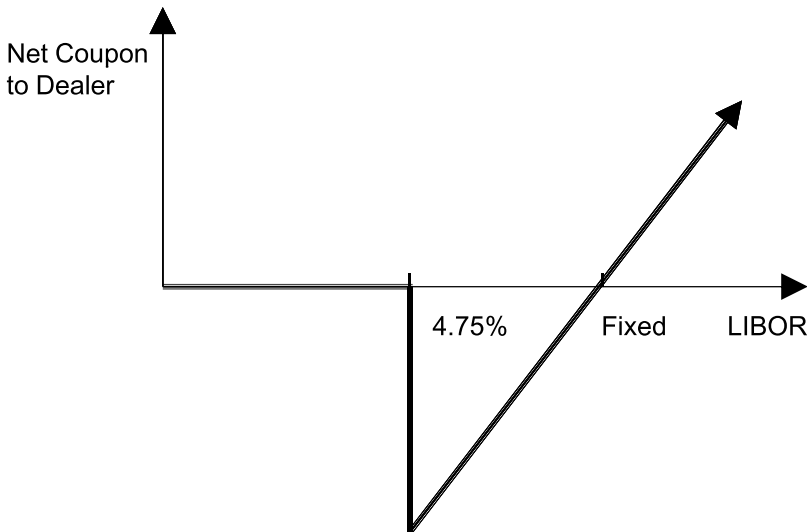


Figure 6.14: Net Coupon Paid Each of the Last Three Years as a Function of LIBOR in Period-by-Period Knockout Swap

in Table 6.7. Notice that this table indicates that these three components amount to the dealer paying fixed and receiving LIBOR when LIBOR is greater than 4.75%, and otherwise the coupons knock out.

Each of these components is easy to value. The present value of the cash flow received at time t_{i+1} from paying fixed in the swap is

	LIBOR $\leq 4.75\%$	LIBOR $> 4.75\%$
Pay Fixed in Swap	LIBOR – Fixed	LIBOR – Fixed
Long Floor	4.75% – LIBOR	0
Long Binary	Fixed – 4.75%	0
Net	0	LIBOR – Fixed

Table 6.7: Payout Table Showing that the Knockout Swap, from the Perspective of the Dealer, Is a Package Consisting of Paying Fixed in a Plain Vanilla Swap, a Long Position in a Floor, and a Long Position in a Binary Floor

$$\frac{\text{Notional}}{[1 + \text{zc}(t_{i+1})]^{t_{i+1}}} \times [f(t_i, t_{i+1}) - \text{Fixed}],$$

where Fixed is equal to the fixed coupon in the knockout swap that we are solving for and $f(t_i, t_{i+1})$ is the forward LIBOR rate that sets at time t_i .

We can use Black's formula in Equation 4.15 to value the floorlet that resets at time t_i and pays off at time t_{i+1} :

$$\frac{\text{Notional}}{[1 + \text{zc}(t_{i+1})]^{t_{i+1}}} \times [XN(-d_2) - f(t_i, t_{i+1})N(-d_1)], \quad (6.3)$$

where

$$d_1 = \frac{\ln [f(t_i, t_{i+1})/X] + \sigma_{t_i}^2 t_i/2}{\sigma_{t_i} \sqrt{t_i}} \quad (6.4)$$

and

$$d_2 = d_1 - \sigma_{t_i} \sqrt{t_i} \quad (6.5)$$

and X is the knockout trigger of 4.75%. Finally, the value of the binary floor that pays out the quantity (Fixed – X) at time t_{i+1} if LIBOR at time t_i is less than X is

$$\frac{\text{Notional}}{[1 + \text{zc}(t_{i+1})]^{t_{i+1}}} \times N(-d_2) \times [\text{Fixed} - X],$$

with d_2 as given in Equation 6.5. In order to find the fixed coupon in the swap, we need to find the fixed coupon that sets the net present value of the

Time	zc	Forward	d_1	d_2
1	5.000%	5.000%	n/a	n/a
2	5.113%	5.226%	0.5773	0.3773
3	5.312%	5.712%	0.7936	0.5107
4	5.494%	6.041%	0.8674	0.5210
Time	PV Fixed	PV Float	Floor	Binary
1	(5,628,218)	4,761,905	n/a	n/a
2	(5,348,708)	4,729,739	184,329	370,482
3	(5,059,689)	4,890,656	194,210	302,586
4	(4,771,429)	4,877,792	214,358	281,988
	(20,808,044)	19,260,092	592,896	955,056
			NPV	0
Vol	20%			
Knockout	4.75%			
Notional	100,000,000			
Fixed rate	5.910%			

Figure 6.15: Solving for the Fixed Coupon in the Knockout Swap

sum of these three component positions to zero. Figure 6.15 shows how this is done by iterating for the solution in a spreadsheet.

As shown in this figure, the fixed coupon is 5.91%, a pickup of $5.91\% - 5.47\% = 44$ bps over the 4-year swap rate. The fixed rate receiver in this knockout swap is taking the view that LIBOR will not set below 4.75% in any period. Although low LIBOR settings typically benefit a fixed rate receiver, in this trade he is willing to forgo a high net payment in the instances in which LIBOR sets below the knockout trigger. As compensation for accepting this risk, he receives a fixed rate that is 44 bps above the plain vanilla 4-year swap rate.

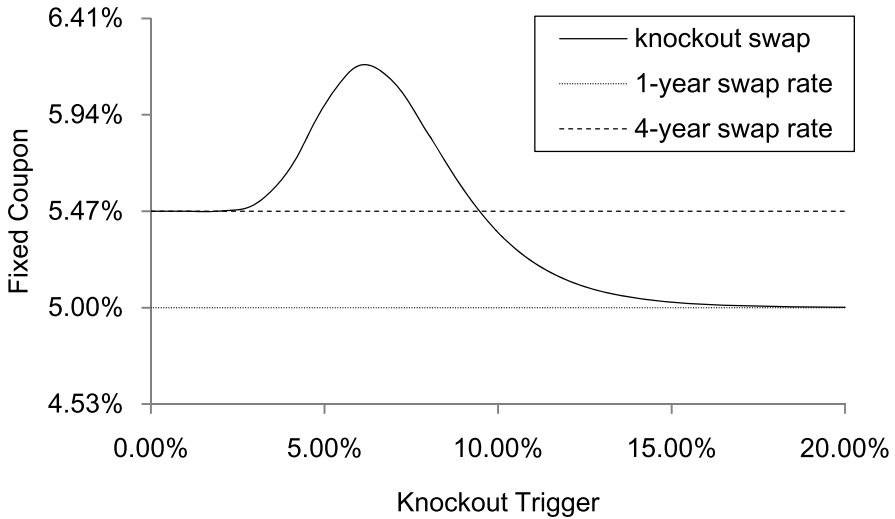


Figure 6.16: Fixed Coupon in the Knockout Swap as a Function of the Knockout Trigger

Figure 6.16 shows how the fixed coupon in the knockout swap varies with the level of the knockout trigger. If the knockout trigger is very high, then there is a very high probability that each of the three LIBOR settings in the back three years will fall below the knockout trigger, in which case the swap will knock out in each year starting in one year.³³ That is, for very high levels of the knockout trigger, it is a virtual certainty that the swap will only pay coupons in one year based off of today's LIBOR setting. Thus, we would expect the coupon on the knockout swap to be very close to today's 1-year swap rate of 5%, which it is.

Conversely, as we move the knockout trigger lower and lower, we decrease the probability that any of the LIBOR settings over the next three years will be below the knockout trigger. That is, for extremely low levels of the knockout trigger, there is a very high probability that the swap will not knock out in any period, in which case the swap will behave very much like a plain vanilla 4-year swap. In this case, we would expect the coupon on the knockout swap to be very close to today's 4-year swap rate of 5.47%, which it is.

³³In this figure we assume that coupon payments will be made at the end of the first year regardless of the level of the knockout trigger.

For intermediate values of the knockout trigger, it is more uncertain whether or not any of the periods of the swap will knock out. In these cases, the investor who receives fixed in the knockout swap is compensated for this risk by receiving a higher coupon. A derivative structurer will try to come up with a knockout trigger that provides a high coupon and also fits with the view of the investor. $\square$

6.5 Swaps with Convexity Adjustments

A *convexity adjustment* is a somewhat subtle concept to many in the industry. There are a variety of situations in which a convexity adjustment comes into play in the swaps market.³⁴ We have already seen an example of a situation that requires a convexity adjustment, namely, the use of Eurodollars to price a swap as we discussed in Chapter 3. In another example, when a swap has in its terms a linear payout of a rate that otherwise defines an instrument that exhibits convexity, it is necessary to make an adjustment to the rate that is implied by the forwards. For example, suppose we enter into a swap in which one counterparty pays a fixed rate and the other pays 10-year CMS. In this case, the swap involves a linear payout of 10-year CMS: for every bp that 10-year CMS moves, the payment in the swap moves a bp.

But 10-year CMS is by definition a floating rate of interest equal to the 10-year swap rate as measured at a certain time each day. As we discussed in Chapter 3, a swap has convexity. Thus, whereas a 1 bp move in the 10-year swap rate creates a 1 bp move in 10-year CMS, the impact to the value of the underlying 10-year swap itself changes as rates move as a function of the convexity of this instrument. The upshot here is that this creates a situation in which convexity adjustments (a change or adjustment made to the pricing of the swap due to the convexity of the underlying instrument) is necessary. This existence of convexity adjustments is often difficult to see and at this point is better seen by working through a few examples.

6.5.1 LIBOR in Arrears Swaps

The popularity of LIBOR in arrears swaps has waxed and waned over the years. LIBOR in arrears swaps were extremely popular in the early 1990s. But then the trade fell out of favor after Orange County's investment portfolio went bankrupt in 1994 due to their use of derivatives.³⁵ Few wanted to

³⁴For further details on the pricing of various swaps with convexity adjustments, see, for example, Hull (2008), Pugachevsky (2001), and Hagan (2003).

³⁵Orange County invested in bonds whose performance was linked to highly sophisticated and highly leveraged derivatives. These derivative trades performed poorly for

do a trade that was funky in the years following Orange County. But as time passed, people were again willing to venture out away from the most plain vanilla trades. And in 2000, LIBOR in arrears swaps were a popular trade once again.

Recall that the standard convention in the swaps market is that LIBOR “sets in advance, pays in arrears,” meaning that LIBOR is set at the beginning of each period and accrues and is paid at the end of the period. In a LIBOR in arrears swap, the LIBOR leg both sets and pays in arrears. This means that the LIBOR rate that is paid each quarter sets at the end of the quarter instead of at the beginning. The accrual period is exactly the same in each case, but the LIBOR rate that is accrued differs. This difference is shown in Figure 6.17.

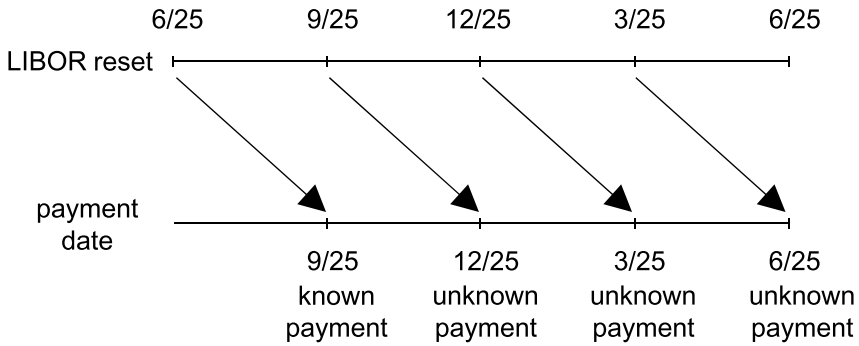
When we think about pricing up a LIBOR in arrears swaps, the floating flows are going to be based off of LIBOR forwards set in arrears. This is demonstrated in Figure 6.18. This figure is drawn with our stylized example and the swap curve in Table 6.1 in mind. This curve is upward sloping, and the LIBOR forwards are upward sloping as well. Let’s try to shed some light on this figure. Suppose that in our stylized example we bootstrap the curve and obtain forward LIBOR rates. The first three LIBOR forwards will be used to price up a plain vanilla 3-year swap. These are the forwards that are indicated by an “ $\times$ ” in Figure 6.18. The fixed rate is set such that this is an NPV = 0 transaction. That is, the fixed rate is set such that the present value of the fixed flows equals the present value of the floating flows that are based off of these LIBOR forwards.

But with LIBOR in arrears we will use a different set of LIBOR forwards to price up the swap. Compared to a plain vanilla swap, an arrears swap will grab the next successive LIBOR forward in each period in order to price up the swap. The three LIBOR forwards that will be used to price up a 3-year swap with LIBOR in arrears are indicated by a “ $\circ$ ” in Figure 6.18.

Let’s make a few observations in Figure 6.18 about how the pricing of an arrears swap differs from a regular swap. First, the LIBOR in arrears trade does not make use of the 0x1 LIBOR forward (which is of course today’s spot 1-year LIBOR rate) to price the swap. It does, however, make use of an additional forward rate (represented by the rightmost $\circ$) in the last period that the regular swap does not use. When the forward curve is upward sloping, this means that the arrears swap will drop a relatively low forward and replace it with a higher forward. In fact, in each period, the forward used to price the arrears swap is higher than the forward used to price a regular swap. The two swaps use some of the same forward rates to

Orange County, and as a result their investment portfolio had to declare bankruptcy in 1994. We’ll talk a lot about the bankruptcy of Orange Country in Chapter 7.

Standard Reset and Payment Conventions



LIBOR in Arrears Reset and Payment Conventions

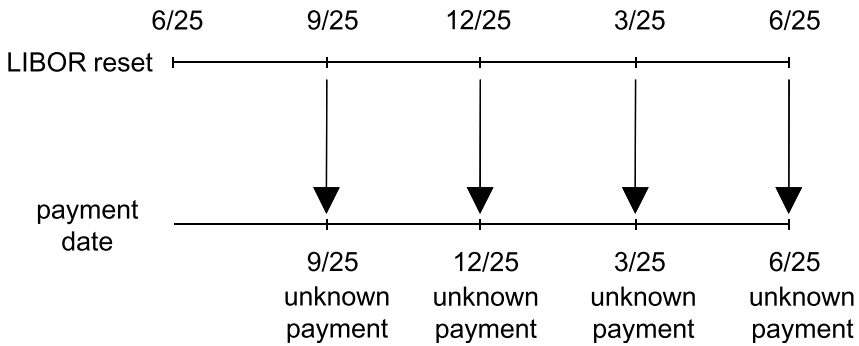


Figure 6.17: Comparison of LIBOR Settings in Standard Swap Versus LIBOR in Arrears Swap

price, but when the curve is upward sloping, the arrears swap will “pay” higher forward rates a period before the regular swap will.

All of these reasons suggest that when the forward curve is upward sloping, the fixed rate paid in a LIBOR in arrears swap should be higher than the fixed rate paid in a regular swap. Let’s take a look now at our stylized example and the swap curve in Table 6.1 and model up a 3-year swap with LIBOR in arrears. This is shown in Figure 6.19. The column labeled “LIBOR in Arrears” shows the three forwards that are used to price

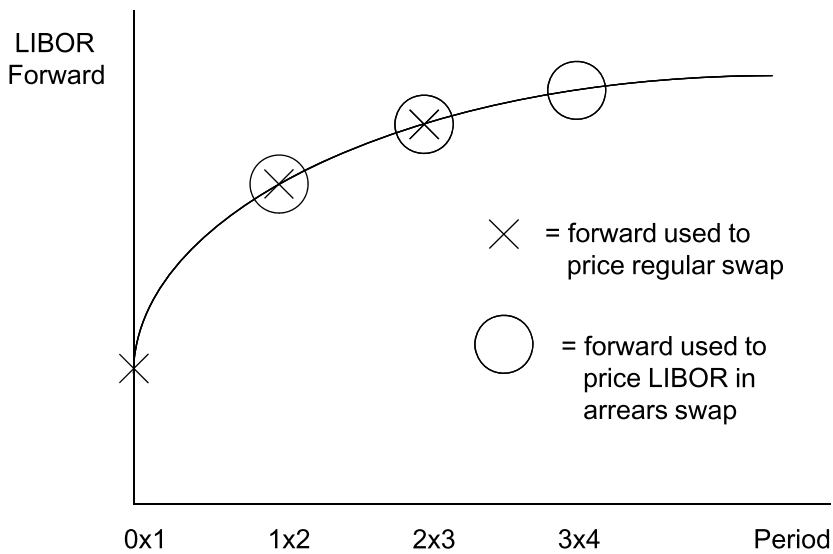
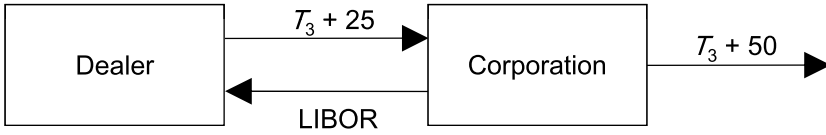


Figure 6.18: Forwards Used in Pricing a 3-Year LIBOR in Arrears Swap Versus a 3-Year Plain Vanilla Swap

LIBOR							
Time	zc	Forward	in Arrears	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	5.226%	5,645,330	5,376,504	5,225,748	4,976,903
2	5.113%	5.226%	5.712%	5,645,330	5,109,495	5,712,202	5,170,021
3	5.312%	5.712%	6.041%	5,645,330	4,833,401	6,041,364	5,172,477
4	5.494%	6.041%		0	0	0	0
					15,319,401		15,319,401
Fixed rate				5.645%	NPV	0	
Notional				100,000,000			

Figure 6.19: Pricing a 3-Year LIBOR in Arrears Swap

Plain Vanilla Swap



LIBOR in Arrears Swap

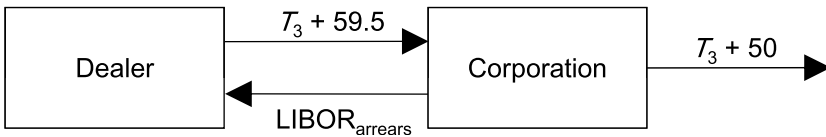


Figure 6.20: Comparison of Corporation Issuing Fixed and Swapping to LIBOR Using Standard Swap and LIBOR in Arrears Swap

the swap. Floating cash flows are determined from these three forwards and are discounted back using the appropriate zero coupon rates.³⁶ The fixed rate in the swap is that rate that makes the swap an $\text{NPV} = 0$ transaction and is 5.645%.

The fixed rate of 5.645% in the 3-year LIBOR in arrears swap is 34.5 bps higher than the 3-year swap rate of 5.30%. For a fixed rate receiver, this is the attractive feature of LIBOR in arrears swaps. Assuming the curve is upward sloping, a counterparty who is interested in receiving fixed in a swap can receive a significantly higher fixed rate if he is willing to pay LIBOR in arrears. The steeper the swap curve, typically the greater the pickup in the fixed rate. In the early 1990s LIBOR in arrears swaps were extremely popular with corporations who issued fixed rate bonds and wanted to swap these liabilities to floating. By paying LIBOR in arrears, these corporations received higher fixed rates that (seemingly) improved their synthetic floating rate funding by the amount of the pickup in the arrears swap. Figure 6.20 demonstrates this in the context of the arrears swap we priced up in Figure 6.19. As shown in this figure, the corporation is assumed to issue a 3-year fixed rate bond at $T_3 + 50$ bps. If it enters into a plain vanilla swap, it receives fixed at $T_3 + 25$

³⁶Note that although this swap has LIBOR set in arrears, this does not mean that we use different zero coupon rates to compute the present value of cash flows. The present value of each cash flow is determined using the appropriate zero coupon rate as we always have, found in the same row as the nominal cash flow in the model.

(the 3-year swap spread in our stylized example is 25 bps, as shown in Table 6.1). This creates synthetic floating rate funding of $\text{LIBOR} + 25$ bps for the corporation. Alternatively, if the corporation enters into a LIBOR in arrears swap, it picks up an additional 34.5 bps on the fixed rate coupon. That is, the corporation will receive $T_3 + 59.5$ bps versus paying LIBOR in arrears. This means that against its issuance at $T_3 + 50$, the corporation has created synthetic floating rate funding of $\text{LIBOR}_{\text{arrears}} - 9.5$ bps.

The synthetic floating rate funding obtained by the corporation with a LIBOR in arrears swaps does appear optically better, but this is at the expense of taking on different risk relative to entering into a plain vanilla swap. If LIBOR rates increase each period over the life of the swap, the corporation winds up paying higher floating rates each period than if they had entered into a plain vanilla swap.³⁷

Although LIBOR in arrears swaps were very popular in the mid-1990s, their use nearly vanished by 1995. There were a couple of reasons for this. First, as mentioned after the bankruptcy of Orange County's investment portfolio (which we will discuss in Chapter 7) due in part to their use of complex derivatives, most investors shied away from anything but the most plain vanilla of derivative products. Second, we just discussed that although the higher fixed rate in LIBOR in arrears swaps makes synthetic floating rate funding appear more attractive, the floating rate payments made in LIBOR in arrears swaps can be dramatically higher in a given quarter if LIBOR rates do increase over the life of the swap.³⁸ This is precisely what happened when the Fed started to raise rates in 1994. Corporations that had entered into arrears swaps, thinking that they had improved their funding costs dramatically due to the pickup in the fixed rate as compared to a plain vanilla swap, watched nervously each quarter as they saw LIBOR rates go higher and higher. Appetite for the product diminished to some extent after this.

Since this time, the use of LIBOR in arrears swaps has been sporadic. During periods of time when the curve is steep (and, as we will see shortly, when implied vol is high), the pickup obtainable in a LIBOR in arrears swap can be significant, and interest in the product increases. But when the curve is relatively flat and implied vol is low, there is little pickup and therefore little interest in LIBOR in arrears swaps.

³⁷ As mentioned, any LIBOR reset in a plain vanilla swap is also a LIBOR reset in the arrears swap, with the exception of the first one. It is a question of timing as to when the resets are paid. And of course the arrears swap also picks up an additional setting at the end of the swap that the plain vanilla swap would not have.

³⁸ Some may consider this risk slightly overstated since each and every LIBOR setting in an arrears swap corresponds with a setting in a corresponding plain vanilla swap, with the exception of the last one.